SESSION 2

WEEK 3

Power Query Editor

Basic Transformations, Data Shaping & Cleaning

Corporate Training · Data Analytics Series · Power BI Foundations & Basics — Module 3

Cross-ref → W10 for Python-based cleaning | Prerequisite: S1 Power BI Desktop Tour

Transform

Clean

Shape

Merge

Load

SESSION 2 — AGENDA & LEARNING OBJECTIVES

90 min

01 What is Power Query?

Role in Power BI, how it fits in the ETL pipeline, M language basics

10 min

02 Power Query Editor Interface

Ribbon, Query Pane, Applied Steps, Formula Bar, Data Preview

15 min

03 Basic Transformations

Change types, rename, reorder, remove columns; filter & sort rows

20 min

04 Data Shaping

Split columns, merge columns, pivot/unpivot, group by, append

20 min

05 Data Cleaning

Remove duplicates/blanks, trim, replace values, fill down/up, errors

15 min

06 Project — Retail Sales Dataset

End-to-end cleaning & shaping exercise on messy real-world data

10 min

WHAT IS POWER QUERY? — THE ETL ENGINE OF POWER BI

Power Query is Power BI's built-in ETL (Extract → Transform → Load) engine. Every step you apply is recorded as M-language code and replays automatically on each data refresh.

EXTRACT

Connect to 150+ sources
Excel, CSV, SQL, APIs,
SharePoint, Cloud...

TRANSFORM

Clean, shape & reshape
Filter, rename, split,
pivot, merge, group...

LOAD

Load clean data into
Power BI Data Model
ready for DAX & visuals

M Language

Every step = M code. View it in Advanced Editor. No need to write it manually — GUI builds it for you.

Auto-Refresh

All steps replay on refresh. Fix source data once — Power Query handles the rest automatically.

Non-Destructive

Never modifies your source file. All transforms happen in Power BI only. Original data is always safe.

POWER QUERY EDITOR — INTERFACE WALKTHROUGH

How to Open: Home ribbon → Transform Data (or double-click any table in Data View)

① RIBBON — Home | Transform | Add Column | View | Tools | Help

② QUERIES PANE

Lists all loaded tables

Right-click to manage

OrderID	Date	Customer	Product	Amount	Status

③ DATA PREVIEW
(shows first 1,000 rows — scroll to inspect)

④ APPLIED STEPS

- Source
- Navigation
- Changed Type
- Removed Cols
- ▶ Filtered Rows
- Renamed Cols

⑤ FORMULA BAR → `= Table.SelectRows(#"Renamed Cols", each [Status] = "Completed")`

💡 Delete any step to undo that transform

⑥ STATUS BAR — Shows column count, row count, last refresh time & any errors detected

BASIC TRANSFORMATIONS — COLUMNS

Home & Transform tabs

🔢 Change Data Type

- Select column(s) → click data type icon in column header
- OR: Home → Data Type dropdown
- Types: Whole Number, Decimal, Text, Date, Date/Time, True/False, Duration

💡 Always fix data types FIRST — wrong types cause filter and calculation errors downstream.

✏️ Rename Columns

- Double-click column header → type new name → Enter
- OR: Right-click header → Rename
- Use clear names: 'OrderDate' not 'Column3' or 'dt_ord'

💡 Rename after 'Changed Type' step so Applied Steps stay readable.

🗑️ Remove Columns

- Select column(s) (Ctrl+click for multiple) → Home → Remove Columns
- OR: Right-click header → Remove
- 'Remove Other Columns' — keeps ONLY selected cols (safer for wide tables)

💡 Use 'Remove Other Columns' when source adds new columns — avoids step errors.

↔️ Reorder Columns

- Drag column headers left/right to reorder visually
- OR: select column → Transform → Move → First/Last/Before/After
- Column order in PQ = column order in Data Model

💡 Put key ID columns first — makes Data View and DAX measures easier to navigate.

🔍 Filter Rows

- Click dropdown arrow on column header → choose filter criteria
- Home → Keep Rows / Remove Rows for preset options
- Combine multiple column filters — each adds an AND step

💡 Filter early in Applied Steps — reduces rows and speeds up all downstream steps.

🔃 Sort Rows

- Click column header dropdown → Sort Ascending/Descending
- Home → Sort Ascending or Sort Descending buttons
- Note: Sort in PQ only matters if row ORDER matters for Fill Down operations

💡 For report sorting, sort in Report View (visual-level). PQ sort = data order in model.

DATA SHAPING — RESTRUCTURING YOUR DATA

Transform tab

✂️ Split Column

Split one column into multiple by delimiter (comma, space, slash) or number of characters.

```
eg: 'John Smith' → 'John' | 'Smith'
'2024-01-15' → '2024' | '01' | '15'
```

→ Select column → Transform → Split Column → By Delimiter or By # of Characters

🔗 Merge Columns

Combine two or more columns into one with a separator character.

```
eg: 'First' + ' ' + 'Last' → 'Full Name'
'City' + ', ' + 'Country' → 'Location'
```

→ Select columns (Ctrl+click) → Transform → Merge Columns → choose separator

🔄 Pivot Column

Turn row values into column headers — converts long format to wide format.

```
eg: Product | Month | Sales
Apple | Jan | 500  →  Jan|Feb|Mar as cols
```

→ Transform → Pivot Column → choose Values Column & Aggregate function

⬇️ Unpivot Columns

Turn column headers into row values — opposite of Pivot. Very common for Excel reports.

```
eg: Jan|Feb|Mar headers → single 'Month' column
(wide format → long format for Power BI)
```

→ Select columns to unpivot → Transform → Unpivot Columns

📦 Group By

Aggregate rows by one or more columns — like SQL GROUP BY. Creates summary tables.

```
eg: Group by Region → Sum of Sales
Group by Customer → Count of Orders
```

→ Home → Group By → select group column + aggregation (Sum, Count, Average, Max...)

📎 Append Queries

Stack tables on top of each other (like UNION in SQL). Columns matched by name.

```
eg: Sales_Jan + Sales_Feb + Sales_Mar
→ Sales_All (full year combined)
```

→ Home → Append Queries → select tables to stack. Use 'Append as New' to keep originals

DATA CLEANING — FIXING MESSY DATA IN POWER QUERY

ROW-LEVEL CLEANING

Remove Duplicates

Select key column(s) → Home → Remove Rows → Remove Duplicates

💡 Use on a unique ID column for exact dedup

Remove Blank Rows

Home → Remove Rows → Remove Blank Rows

💡 Removes rows where ALL cells are empty

Remove Errors

Home → Remove Rows → Remove Errors

💡 Only after inspecting — you may want to fix not delete

Keep/Remove Top N

Home → Keep Rows → Keep Top Rows (enter N)

💡 Useful to strip metadata rows at top of exported files

VALUE-LEVEL CLEANING

Trim (Whitespace)

Transform → Format → Trim

💡 Removes leading & trailing spaces — CRITICAL for lookups/merges

Replace Values

Transform → Replace Values → enter old/new value

💡 Also replaces errors: Transform → Replace Errors

Fill Down / Fill Up

Select column → Transform → Fill → Down or Up

💡 Fills null cells with nearest non-null value above/below

Clean (Non-Printable)

Transform → Format → Clean

💡 Strips invisible characters from system exports

⚠️ Cleaning ORDER matters: Fix types → Trim/Clean → Remove duplicates → Replace values → Fill down. Wrong order = wrong results!

APPLIED STEPS & THE M LANGUAGE — UNDER THE HOOD

HOW APPLIED STEPS WORK

1 **Source**
`Excel.Workbook(File.Contents("Sales.xlsx"))`

2 **Navigation**
`Source{[Item="Sales_Table"]}[Data]`

3 **Changed Type**
`Table.TransformColumnTypes(...)`

4 **Removed Columns**
`Table.RemoveColumns(...,{"Col1","Col2"})`

5 **Filtered Rows**
`Table.SelectRows(...,each [Status]="Active")`

6 **Renamed Columns**
`Table.RenameColumns(...,{{"dt","Date"}})`

🔑 Click any step to see data state at THAT point. Delete a step to undo. Drag steps to reorder (careful — may break dependencies).

M LANGUAGE — ADVANCED EDITOR EXAMPLE

```
let
  Source = Excel.Workbook(
    File.Contents("Sales.xlsx")),
  Sales_Table = Source{[Item=
    "Sales_Table"]}[Data],
  #"Changed Type" =
    Table.TransformColumnTypes(
    Sales_Table,{{"Date",type date},
    {"Amount",type number}}),
  #"Filtered Rows" =
    Table.SelectRows(
    #"Changed Type",
    each [Status] = "Active")
in
  #"Filtered Rows"
```

📋 View → Advanced Editor to see full M script
You can copy/paste or hand-edit M code directly

PROJECT EXERCISE

Retail Sales Data Cleaning Project

End-to-end Power Query transformation of a real-world messy retail dataset

📋 SCENARIO

You are a data analyst at GlobalMart Retail. Your manager sends you a raw Excel export from the ERP system with 3 months of sales transactions. The file is messy — wrong types, duplicates, blanks, inconsistent categories and merged columns. Your job is to clean and shape it in Power Query so it's ready for a regional sales dashboard.

2,847	14	6	12
Raw Rows	Columns	Issues to Fix	PQ Steps

🎯 DELIVERABLE: A clean, query-formatted Power BI dataset with correct types, no duplicates, merged region column, and a summary Group By table — ready for DAX measures in S3.

PROJECT — STEP 1: UNDERSTAND THE RAW DATA

GlobalMart Retail Dataset

RAW DATASET — WHAT WE RECEIVE FROM ERP

order_id	DATE	cust name	REGION	sub_region	product	qty	unit_price	discount%	total	status	returned	sales_rep	notes
ORD-001	1/5/2024	john smith	North	north-east	Laptop Pro	2	1299.99	10%	2339.98	completed	No	Sarah J.	bulk
ORD-001	1/5/2024	john smith	North	north-east	Laptop Pro	2	1299.99	10%	2339.98	completed	No	Sarah J.	bulk
ORD-003	5-Jan-24	Maria G.	SOUTH	south_west	Monitor	1	349.50	0%	349.50	Completed	No	Tom B.	
ORD-004		Bob L.	East	east	Keyboard	5	89.99	5%	427.45	pending	Yes		
ORD-005	2024/01/07	Alice W.	west		Mouse	3	29.99	0%	null	COMPLETED	no	Sarah J.	rush

Column with issues

Duplicate row

Blank / null values

6 ISSUES IDENTIFIED — Tasks for Power Query:

1 **Column naming inconsistent**
Mix of UPPERCASE, lowercase, snake_case, spaces — rename all to consistent PascalCase
Rename Columns

2 **DATE column — 3 different formats**
'1/5/2024', '5-Jan-24', '2024/01/07' — Power BI cannot aggregate mixed formats
Change Type → Date

3 **Duplicate rows (ORD-001 appears twice)**
ERP exports often include duplicates on re-exports — must be removed by OrderID
Remove Duplicates

4 **Trailing spaces in customer names**
'john smith ' has trailing space — causes mismatches in lookups and GROUP BY
Trim + Proper Case

5 **REGION & SUB_REGION inconsistent case + separator**
'North', 'SOUTH', 'west' + 'north-east', 'south_west' — merge into single Region column
Replace + Merge Columns

6 **Discount% is text, Total has 'null' as text**
Discount stored as '10%' string — strip %, convert to decimal. 'null' string ≠ blank
Replace + Change Type

PROJECT — STEP 2: POWER QUERY SOLUTION (12 APPLIED STEPS)

Follow Along

01
Source + Navigate
Get Data → Excel → select 'Sales_Raw' table
```
Source → Navigation
```

02
Rename All Columns
Right-click each header → Rename to: OrderID, Date, CustomerName, Region, SubRegion, Product, Qty, UnitPrice, DiscountPct, Total, Status, Returned, SalesRep, Notes
```
Table.RenameColumns(...)
```

03
Change Date Type
Select Date column → Data Type → Date (Power Query auto-normalises all 3 formats)
```
Table.TransformColumnTypes(...,{{"Date",type date}})
```

04
Fix Discount Column
Replace Values: '10%'→'10', '5%'→'5', '0%'→'0' → Change Type → Decimal Number
```
Table.ReplaceValue(...) + TransformColumnTypes
```

05
Fix Total 'null' text
Replace Values: 'null' → null (empty) → Change Type → Decimal Number
```
Table.ReplaceValue(#"Step",'null',null,...)
```

06
Trim Customer Names
Select CustomerName → Transform → Format → Trim then Format → Capitalize Each Word
```
Table.TransformColumns(...,{{"CustomerName",Text.Proper}})
```

07
Standardise Region Case
Select Region → Transform → Format → Capitalize Each Word (fixes SOUTH → South, west → West)
```
Text.Proper([Region])
```

08
Merge Region + SubRegion
Ctrl+click Region, SubRegion → Transform → Merge Columns → Separator: ' – ' → name 'FullRegion'
```
Table.CombineColumns(...,{"Region","SubRegion"})
```

09
Remove Duplicates
Select OrderID column → Home → Remove Rows → Remove Duplicates
```
Table.Distinct(#"Step",{"OrderID"})
```

10
Remove Notes Column
Select Notes column → Home → Remove Columns (not needed for dashboard)
```
Table.RemoveColumns(...,{"Notes"})
```

11
Filter Active Orders
Status column dropdown → Text Filters → Equals 'completed' (case insensitive)
```
Table.SelectRows(...,each Text.Lower([Status])="completed")
```

12
Group By — Regional Summary
Home → Group By → Group by FullRegion → Sum of Total as 'RegionalSales', Count of OrderID as 'OrderCount'
```
Table.Group(...,{"FullRegion"},{...})
```

PROJECT — STEP 3: BEFORE vs AFTER RESULTS

GlobalMart Retail Dataset

❌ BEFORE — Raw Messy Data

order_id	DATE	cust name	REGION	discount%	total	status
ORD-001	1/5/2024	john smith	North	10%	2339.98	completed
ORD-001	1/5/2024	john smith	North	10%	2339.98	completed
ORD-003	5-Jan-24	Maria G.	SOUTH	0%	349.50	Completed
ORD-004		Bob L.	East	5%	427.45	pending
ORD-005	2024/01/07	Alice W.	west	0%	null	COMPLETED

Rows: **2,847**

Duplicates: **143**

Blank dates: **67**

Type errors: **2 cols**

→

✅ AFTER — Clean Shaped Data

OrderID	Date	CustomerName	FullRegion	DiscountPct	Total	Status
ORD-001	05/01/2024	John Smith	North – North-East	0.10	2339.98	completed
ORD-003	05/01/2024	Maria G.	South – South-West	0.00	349.50	completed
ORD-005	07/01/2024	Alice W.	West – West	0.00	(blank)	completed

GROUP BY — Regional Summary Table

FullRegion	RegionalSales	OrderCount
North – North-East	$284,720	412

Clean Rows: **2,704**

Duplicates: **0**

Data Types OK: **✓ All**

Ready for DAX: **✓ Yes**

SESSION 2 — KEY TAKEAWAYS

Power Query is your ETL engine

All transforms are recorded as M code, auto-replay on refresh, and NEVER touch the source file.

Fix data types FIRST

Wrong types cause calculation errors. Always set Date, Number, and Text types before any other operation.

Cleaning order matters

Sequence: Types → Trim/Clean → Remove Duplicates → Replace Values → Fill Down → Filter.

Pivot & Unpivot are power tools

Most Excel reports are wide-format. Unpivot them to long-format for proper Power BI aggregation.

Group By creates summary tables

Use Group By to create aggregate reference tables — perfect for KPI cards and dashboard metrics.

Applied Steps = your audit trail

Every step is named and reversible. Click any step to inspect the data at that point in the pipeline.

NEXT → S3: Building your first report — visuals, slicers & simple relationships | Cross-ref: W10 Python-based cleaning